

SE423 Software Project Management

Activity 7: Risk Identification & Analysis

Group Members:

SHOUG FAWAZ ABDULLAH ALOMRAN

ROSE SAEED RAKAN AL RAKAN

YARA FARIS FAIHAN ALBUGAMI

HATOON ABDULLAH

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1. AI generated document

2. Risks SE423 Risk Activity

1. Scenario

Our software company has been hired to implement a Financial E-Services Platform that provides services to government entities, private-sector organizations, and individuals. The platform aims to enhance partnerships supporting development projects in Saudi Arabia, enable the digital transformation of financial services, improve transparency and efficiency, and facilitate the delivery of these services.

2. AI-Generated System Requirements

2.1 AI Prompt

AI Tool Used: ChatGPT

Prompt:

You are a requirements engineer and have been asked to generate detailed system requirements for Financial E-Services Platform in Saudi Arabia. This system will offer its financial services to both governmental agencies as well as private companies and individuals. The main objective of the system is to support development projects, improve collaboration, enable digital transformation, increase transparency and efficiency, and provide financial services.

Generate detailed functional and non-functional requirements for the system. Be creative but ensure that the requirements are realistic and applicable to a large scale financial platform functioning in Saudi Arabia. Make sure to consider the necessary local regulations and standards for financial services, cybersecurity, financial transaction, privacy etc.

Provide the answer in the format of structured Software Requirements Specification (SRS). Include functional requirements such as user management, financial transaction, partnership & project management, reporting & analytics, notifications & communication etc. Include non-functional requirements like security, performance, availability & reliability, regulation, privacy, scalability, usability, and interoperability.

Provide individual requirements in the form of unique requirement ID and description. Follow the proper requirements-engineering style.

2.2 Generated Requirements

The requirements that have been generated cover the key functionalities and non-functional features of the Financial E-Services Platform. The list of requirements was then used as the basis for risk identification.

2.2.1 Functional Requirements

User Management

- **FR-01:** The system shall support role-based registration for government entities, private-sector organizations, and individual users.
- **FR-02:** The system shall authenticate users through multi-factor authentication (MFA).
- **FR-03:** The system shall enforce role-based access control (RBAC) with configurable permission levels for each user category.
- **FR-04:** The system shall integrate with the National Single Sign-On (SSO) service, Nafath, for identity verification of Saudi nationals.

Financial Transaction Services

- **FR-05:** The system shall enable the initiation, approval, and tracking of fund disbursements for government development projects.
- **FR-06:** The system shall support electronic payment processing, including bank transfers, SADAD payments, and credit/debit card transactions.
- **FR-07:** The system shall generate unique transaction reference numbers and maintain a complete audit trail for every financial operation.
- **FR-08:** The system shall perform real-time validation of transaction data against predefined business rules and regulatory thresholds.

- **FR-09:** The system shall implement automated KYC and AML screening for financial transactions.

Partnership and Project Management

- **FR-10:** The system shall provide a portal for managing development-project partnerships between government and private-sector entities.
- **FR-11:** The system shall track project milestones, deliverables, and associated financial commitments.
- **FR-12:** The system shall generate partnership agreements and contract documents using configurable templates.

Reporting and Analytics

- **FR-13:** The system shall provide configurable dashboards displaying key financial indicators and service metrics.
- **FR-14:** The system shall generate regulatory reports required by supervisory authorities.
- **FR-15:** The system shall support exporting reports in PDF, Excel, and CSV formats.

Notifications and Communication

- **FR-16:** The system shall deliver real-time notifications through email, SMS, and in-application alerts for transaction status changes and approval requests.
- **FR-17:** The system shall provide a secure internal messaging module for communication between platform stakeholders.

These are taken from the SRS your group already generated rather than creating a new set of requirements for the report.

2.2.2 Non-Functional Requirements

Security

- Data at rest shall be encrypted using AES-256, while data in transit shall be protected using TLS 1.2 or higher.
- The system shall apply the principle of least privilege and appropriate access controls.
- The platform shall incorporate intrusion detection and prevention, web application firewall protection, and real-time threat monitoring.
- The system shall undergo periodic vulnerability scanning and penetration testing.

Performance

- The platform shall respond to user requests within **2 seconds under normal load** of up to 5,000 concurrent users and within **5 seconds under peak load** of up to 15,000 concurrent users.
- The platform shall support at least **500 financial transactions per second** during peak operational periods.
- Dashboard and reporting pages shall render within **3 seconds**.

Availability and Reliability

- The system shall maintain at least **99.9% availability**, excluding scheduled maintenance.
- Disaster recovery shall support a **Recovery Time Objective (RTO) of 4 hours** and **Recovery Point Objective (RPO) of 1 hour**.
- Critical components shall support automatic failover.

Data Privacy

- Personal data shall be processed according to applicable data-protection requirements.
- Data shall be classified according to its sensitivity and protected using controls appropriate to its classification.

Scalability

- The architecture shall support horizontal scaling to accommodate growth in users and transaction volumes.
- The system shall use a modular or microservices-based architecture to allow individual services to be independently deployed and scaled.

Usability

- The platform shall support both **Arabic and English**, including right-to-left rendering for Arabic.
- The system shall satisfy applicable accessibility requirements.
- New users shall be able to complete registration and onboarding within 10 minutes without external assistance.

Interoperability

- The system shall provide RESTful APIs for integration with external government systems and private-sector banking platforms.
- The system shall support standardized financial-data exchange formats.

2.2.3 Regulatory and Compliance Requirements

The generated requirements identify compliance as a major consideration because the platform processes financial transactions and sensitive personal information in Saudi Arabia.

- The SRS identifies the following regulatory and compliance considerations: Applicable **Saudi Central Bank (SAMA)** requirements for financial institutions and electronic financial services.
- **National Cybersecurity Authority (NCA)** cybersecurity controls.
- **Saudi Personal Data Protection Law (PDPL)** requirements.
- Applicable **Anti-Money Laundering (AML)** requirements.
- **Know Your Customer (KYC)** controls for financial transactions.

- Data sovereignty requirements requiring relevant financial and personal data to be stored and processed within Saudi Arabia.
- Maintenance of audit logs for user activities and financial transactions.
- Use of approved third-party integrations and vendors where required.
- Operation within the applicable regulatory environment prior to full production deployment.

3. Risk Identification and Analysis

Based on the scenario and the detailed system requirements, project, technical, and business risks were identified and analyzed.

For each identified risk, the following information was determined:

- Risk ID
- Description of Risk
- Risk Category (Project, Technical, or Business)
- Probability of Occurrence
- Impact Loss or Cost (KSAR)
- Risk Exposure (Probability × Impact)
- Main Drivers (Causes)

The complete risk analysis is provided in the accompanying Excel workbook.

3.1 Risk Exposure Calculation

Risk Exposure (RE) was calculated using:

Risk Exposure = Probability × Impact

The identified risks were ranked according to their calculated risk exposure, and the top ten risks were selected.

4. Conclusion

Risk Identification and Risk Analysis techniques were employed to identify the risks that were important with regard to the Financial E-Services Platform. Risks were analyzed according to the likelihood and financial implications associated with the occurrence of risks. The risk exposure levels were used to rank the top ten important risks.